Aim

The primary aim of the FairWater project is to design and implement an intelligent water distribution management system that addresses resource inequality and operational inefficiency by ensuring fair, efficient, and transparent delivery of water across all regions of Raipur.

Core Objectives (Mandate from PS3 Challenge)

1. Operational Monitoring: To develop a real-time monitoring dashboard for municipal authorities, displaying live pipeline pressure, water flow analytics, and per-pipeline diagnostics.

2. Citizen Feedback Loop: To build a public-facing portal for citizens to instantly report shortages or leaks, feeding crucial ground-level data and optional photo evidence directly into the administrative system.

3. AI-Driven Equity: To implement an AI-driven optimization module that automatically adjusts pumping schedules based on demand and live anomaly data, ensuring equitable distribution, particularly for vulnerable "tail-end" zones.

Tools Required

Frontend (Client)

- React 18, TypeScript, Vite
- Tailwind CSS, shadcn/ui
- TanStack Query
- React Leaflet, Recharts

Backend (Server)

- Node.js, Express, TypeScript
- Zod

Data / Infrastructure

- Supabase (PostgreSQL)
- Supabase Storage

Theory

The FairWater system functions as an Adaptive Control System based on the Negative Feedback Loop model to maintain Equitable Water Distribution.

Set Point: Fair pressure/flow across all zones.

Sensor/Input: Live telemetry and citizen reports.

Controller: AI Optimization Logic.

Correction: Boosted Pumping Schedule.

Principle of AI-Driven Equity:

- Detects low-pressure conditions.
- Allocates boosted 40-minute schedules to affected zones.
- Balances others to maintain fairness.

Procedure

Setup & Configuration:

- Set SUPABASE_KEY in .env
- Configure tsconfig.json paths

Data Persistence:

- zone_history and citizen_reports creation in Supabase

Development Cycle:

- pnpm dev for concurrent frontend/backend

Data Generation:

- simulation.ts logs snapshots every 30 seconds

Output

Admin Dashboard (/admin):

- Real-time map, charts, AI scheduling.

Citizen Portal (/):

- Complaint submission with validation.

AI Analysis Tool (/analysis):

- Simulated corrosion detection module.

Result

1. Demonstrates equitable water distribution with AI.

2. Integrates live simulation + citizen reports into Alerts Panel.

3. Strict type safety via TypeScript and Zod.

4. Architecture ready for cloud deployment.

Precautions

System Reliability:

- TanStack Query caching, retries.

Input Integrity:

- Strict Zod validation.

Secret Management:

- Server-only access to SUPABASE_SERVICE_KEY.

Deployment Security:

- RLS enabled on citizen_reports.

Project Code

Key Files:

- server/routes/optimization.ts: AI scheduling logic
- server/routes/simulation.ts: Realtime simulation + logging
- server/routes/report.ts: Citizen report handling
- client/pages/Dashboard.tsx: Admin dashboard UI
- shared/api.ts: Interfaces and contracts